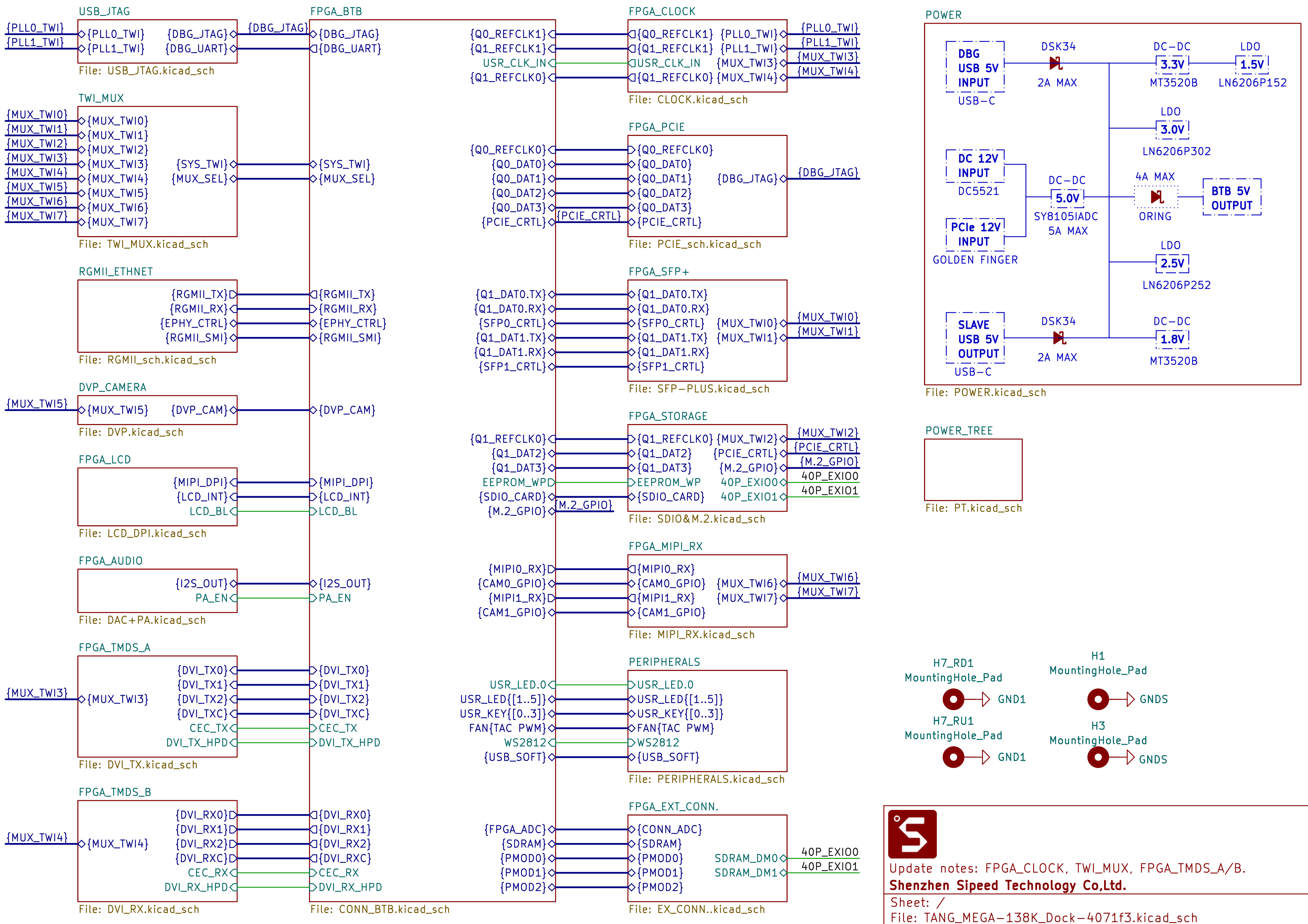


Update notes: FPGA_CLOCK, TWI_MUX, FPGA_TMDS_A/B.

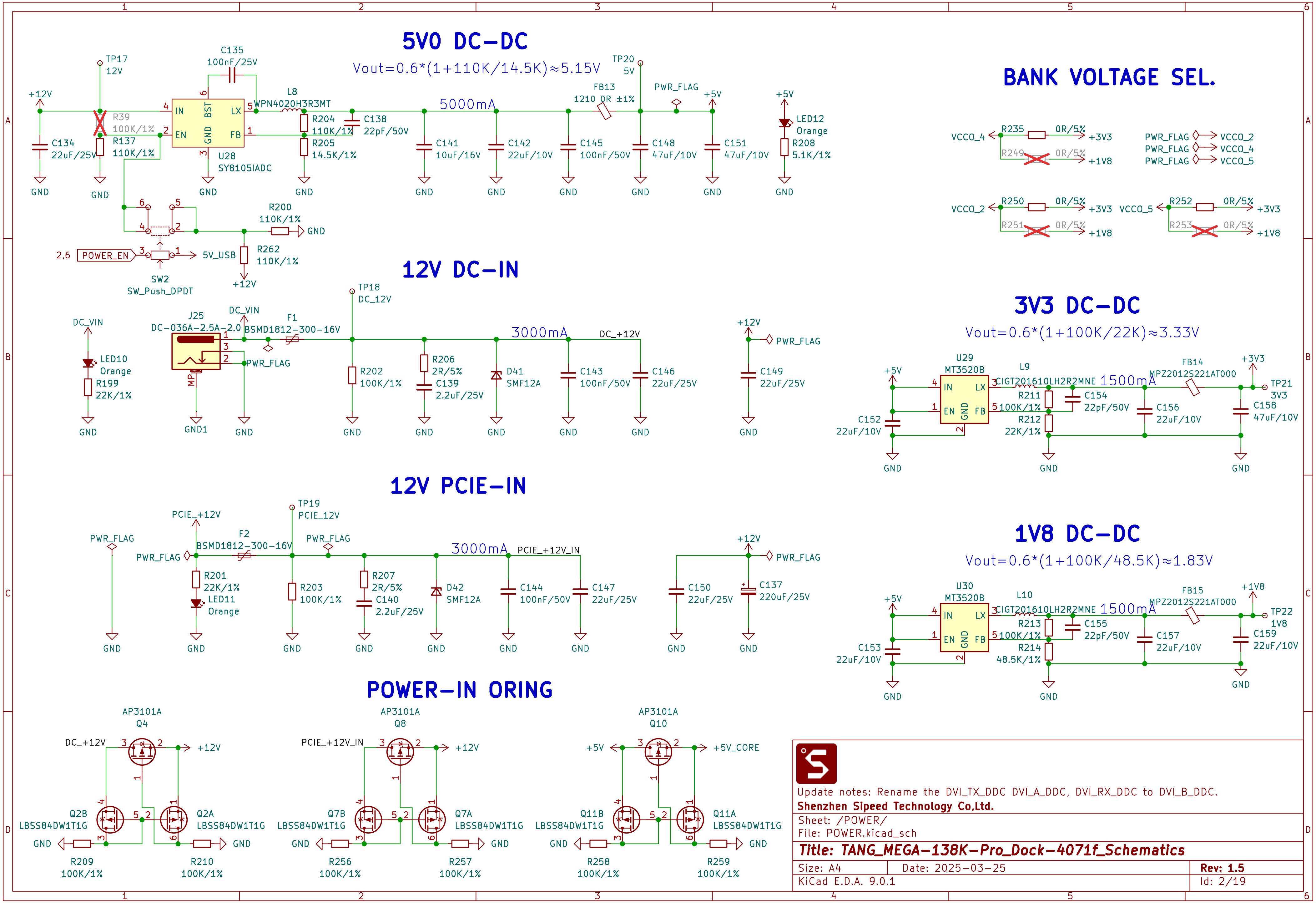
Shenzhen Sipeed Technology Co,Ltd.

Sheet: /

File: TANG_MEGA-138K_Dock-4071f3.kicad_sch

Title: TANG_MEGA-138K-Pro_Dock-4071f_Schematics

Size: A4	Date: 2025-03-25	Rev: 1.5
KiCad E.D.A. 9.0.1		Id: 1/19



Update notes: Rename the DVI_TX_DDC DVI_A_DDC, DVI_RX_DDC to DVI_B_DDC.

Shenzhen Sipeed Technology Co,Ltd.

Sheet: /POWER/

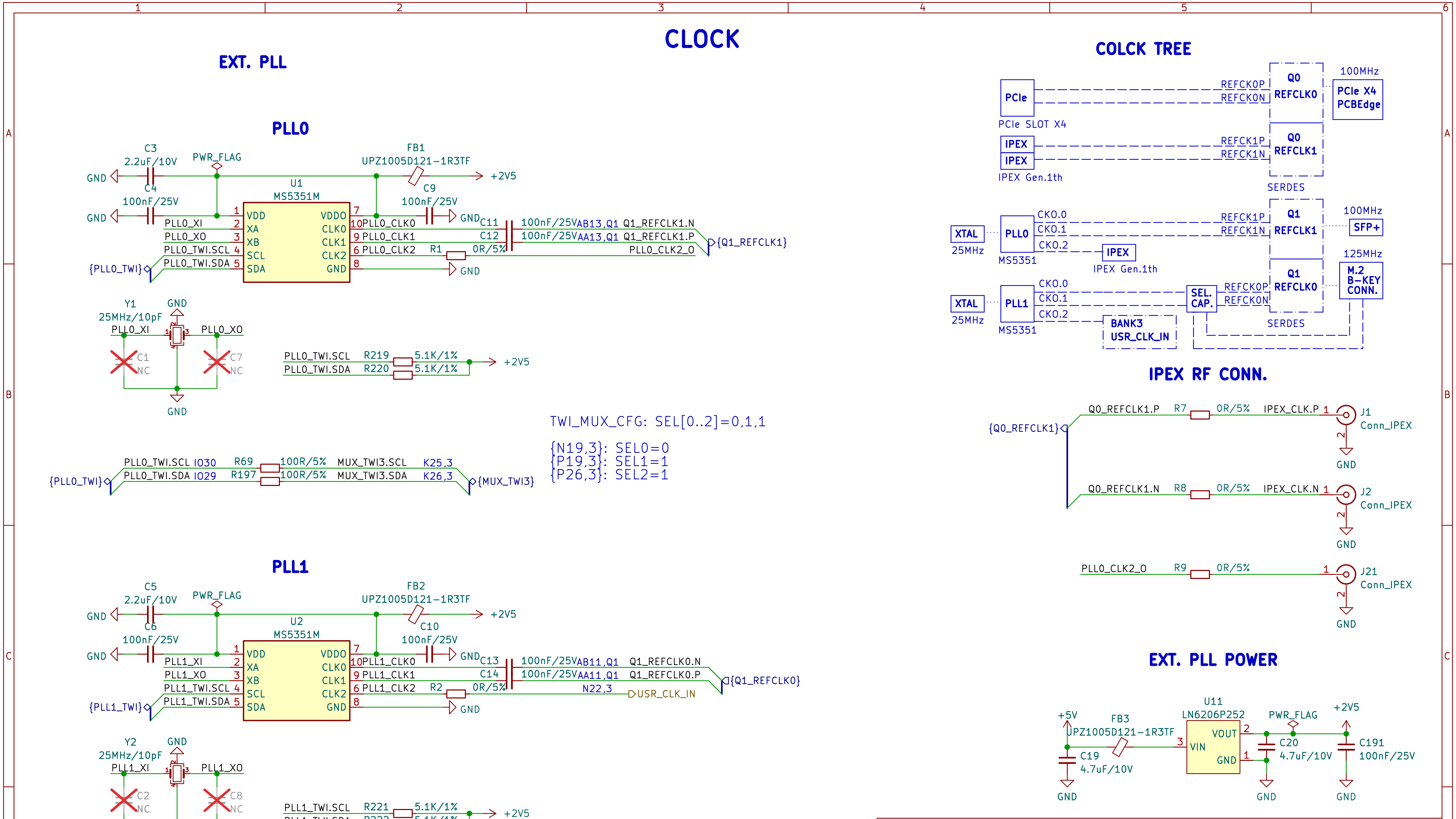
File: POWER.kicad_sch

Title: TANG_MEGA-138K-Pro_Dock-4071f_Schematics

Size: A4 | Date: 2025-03-25 | Rev: 1.5

KiCad E.D.A. 9.0.1 | Id: 2/19

CLOCK



Update notes: Fixed the issues of clock tree.

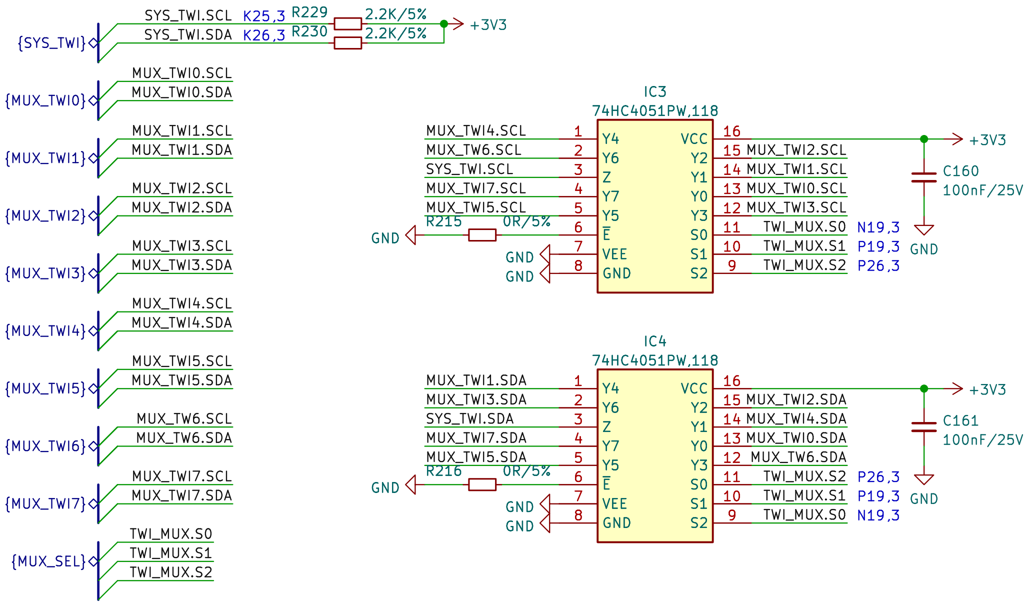
Shenzhen Sipeed Technology Co,Ltd.

Sheet: /FPGA_CLOCK/
File: CLOCK.kicad_sch

Title: TANG_MEGA-138K-Pro_Dock-4071f_Schematics

Size: A4	Date: 2025-03-25	**Rev: 1.5**
KiCad E.D.A. 9.0.1		Id: 3/19

8-WAY TWI MUX



NO.	CONFIG	SEL0	SEL1	SEL2
0	SFP0	0	0	0
1	SFP1	0	0	1
2	EEPROM	0	1	0
3	DVI_A_DDC MS5331-0	0	1	1
4	DVI_B_DDC MS5331-1	1	0	0
5	DVP_CAM	1	0	1
6	MIPI_RX0	1	1	0
7	MIPI_RX1	1	1	1

Update notes: Rename the DVI_TX_DDC to DVI_A_DDC, DVI_RX_DDC to DVI_B_DDC.

Shenzhen Sipeed Technology Co,Ltd.

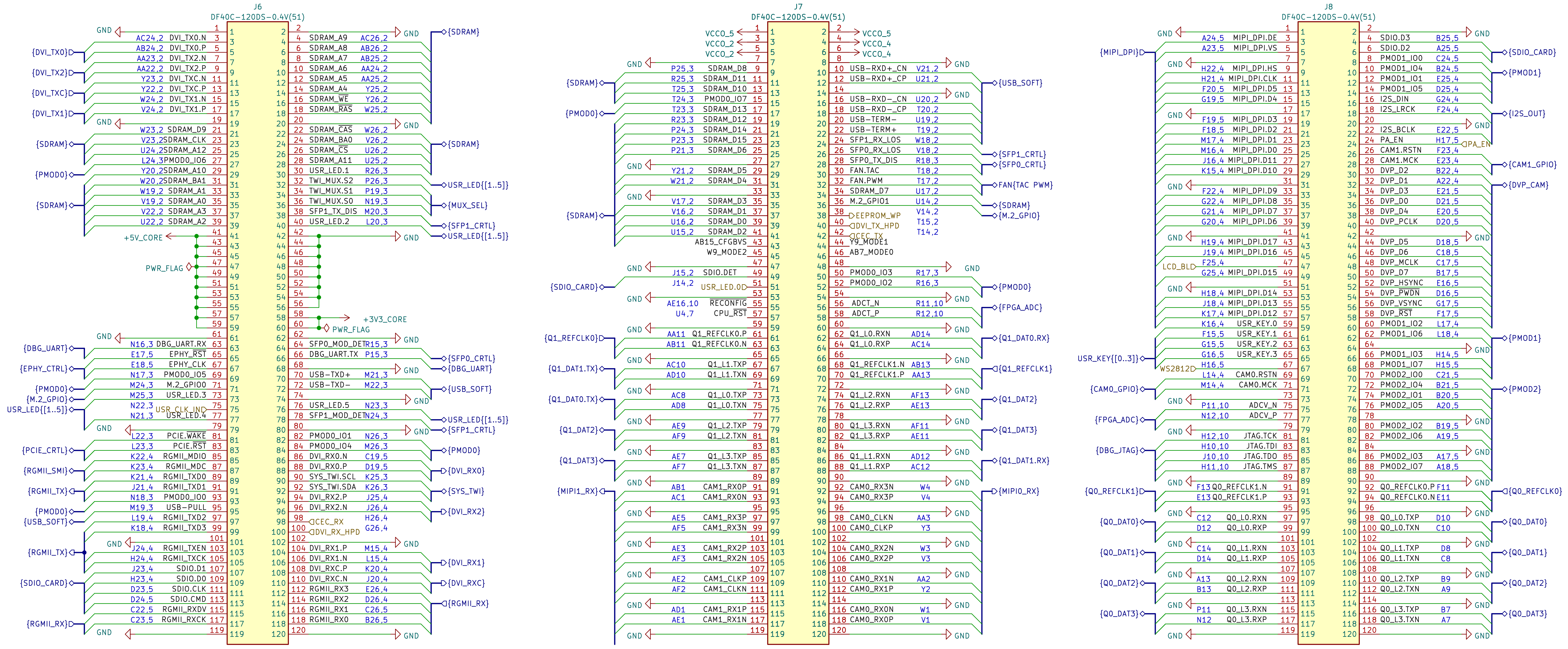
Sheet: /TWI_MUX/
File: TWI_MUX.kicad_sch

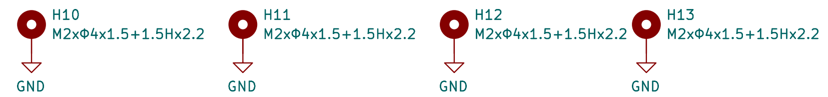
Title: TANG_MEGA-138K-Pro_Dock-4071f_Schematics

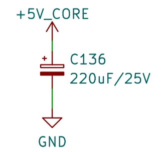
Size: A4 | Date: 2025-03-25 | **Rev: 1.5**
KiCad E.D.A. 9.0.1 | Id: 4/19

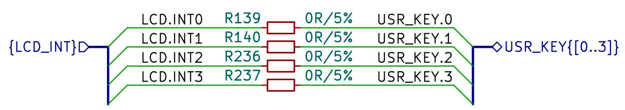
FPGA CORE MODULE

Note: that the pinout (numbers) of these conn. here is mirrored compared to the SOM.

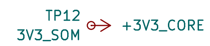




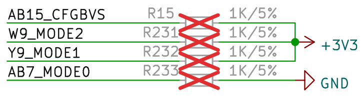




TEST POINT



MODE SET



NOTE: MODE[0..2] is set to 001 by defualt on CM Board
Set CFGBVS to 1 to enable maximum driving of the pin during FPGA configuration

CONFIG	MODE0	MODE1	MODE2
JTAG	X	X	X
MSPI	0	0	1
SSPI	0	1	0
MSERIAL	0	0	0
SSERIAL	1	1	1
MCPU	1	0	0
SCPU	1	1	0

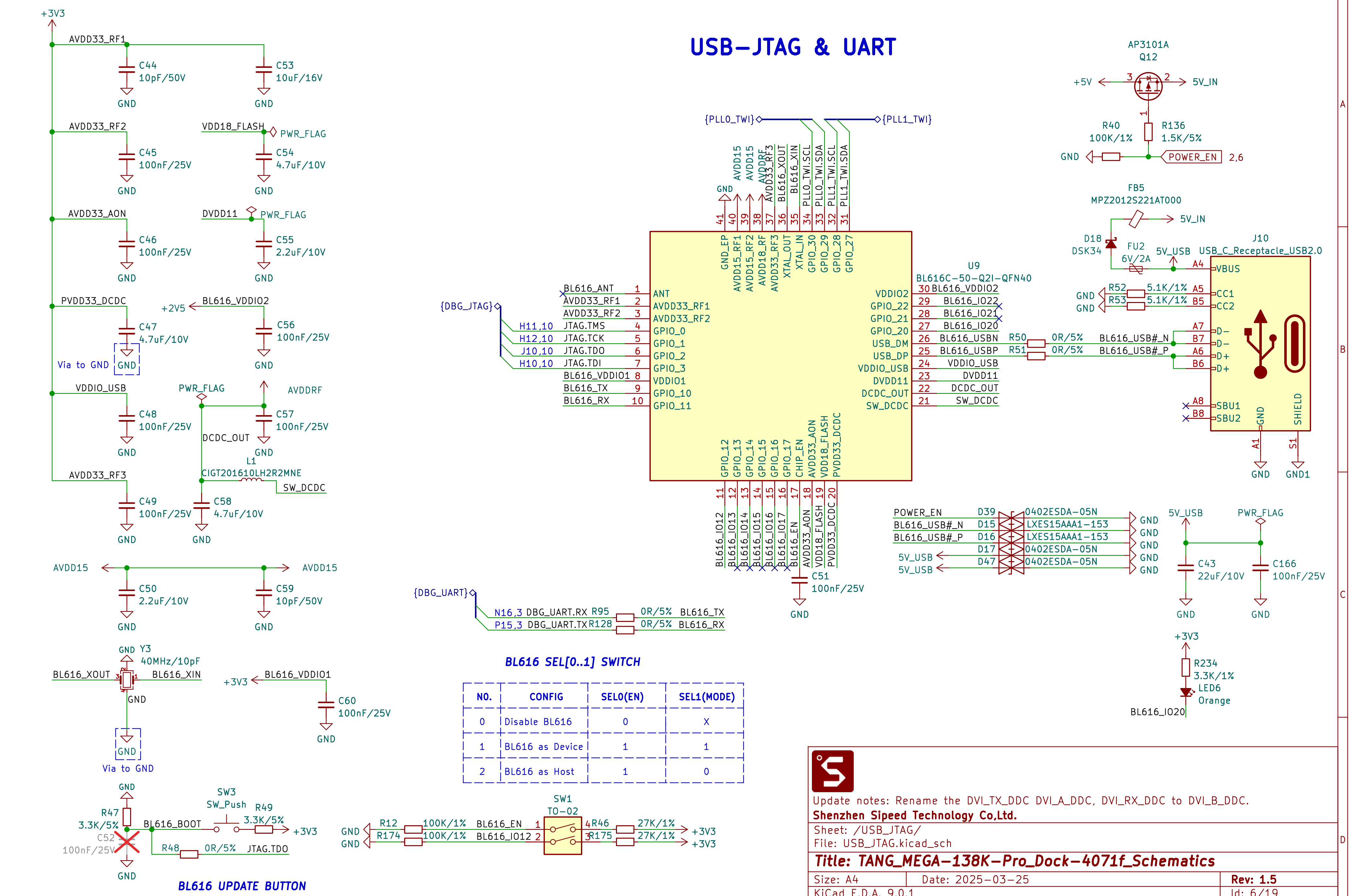
Update notes: Rename the DVI_TX_DDC DVI_A_DDC, DVI_RX_DDC to DVI_B_DDC.

Shenzhen Sipeed Technology Co,Ltd.
Sheet: /FPGA_BTB/
File: CONN_BTB.kicad_sch

Title: TANG_MEGA-138K-Pro_Dock-4071f_Schematics

Size: A3 | Date: 2025-03-25 | Rev: 1.5
KiCad E.D.A. 9.0.1 | Id: 5/19

USB-JTAG & UART



BL616 SEL[0..1] SWITCH

NO.	CONFIG	SEL0(EN)	SEL1(MODE)
0	Disable BL616	0	X
1	BL616 as Device	1	1
2	BL616 as Host	1	0

Update notes: Rename the DVI_TX_DDC DVI_A_DDC, DVI_RX_DDC to DVI_B_DDC.

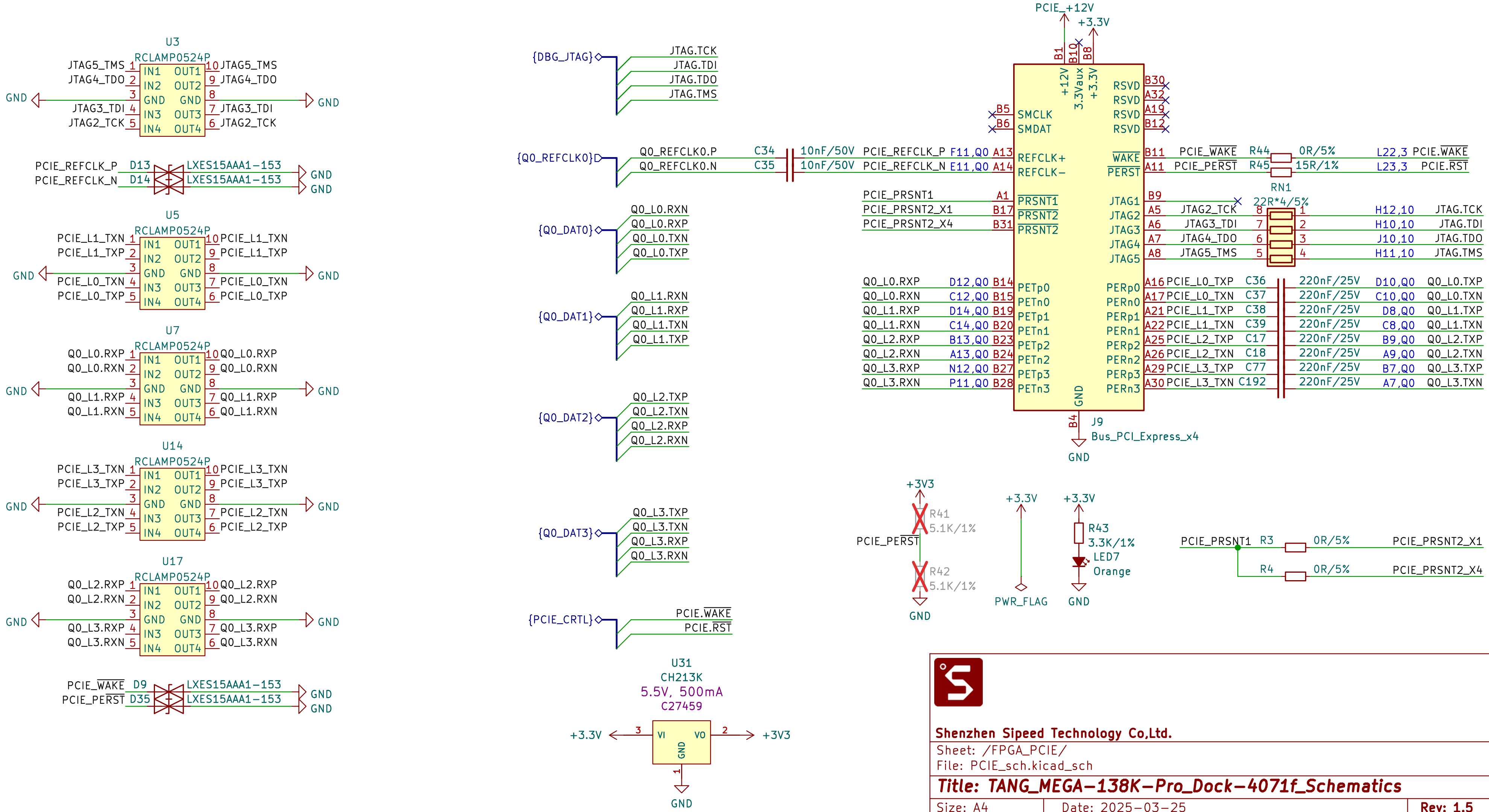
Shenzhen Sipeed Technology Co,Ltd.

Sheet: /USB_JTAG/
File: USB_JTAG.kicad_sch

Title: TANG_MEGA-138K-Pro_Dock-4071f_Schematics

Size: A4	Date: 2025-03-25	Rev: 1.5
KiCad E.D.A. 9.0.1		Id: 6/19

PCI–Express X4



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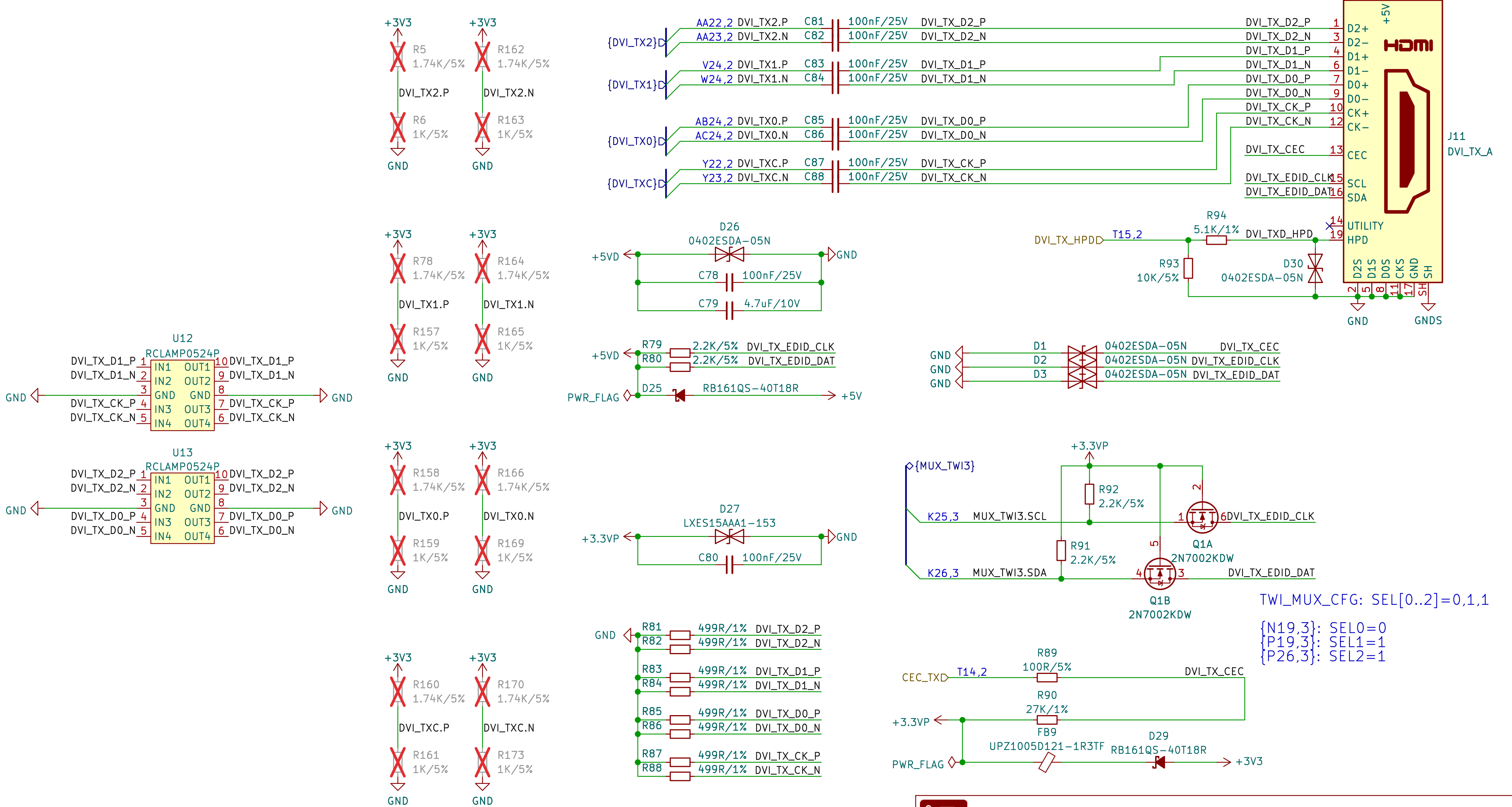
Sheet: /FPGA_PCIE/
File: PCIE_sch.kicad_sch

Title: TANG_MEGA-138K-Pro_Dock-4071f_Schematics

Size: A4 | Date: 2025-03-25 | **Rev: 1.5**

KiCad E.D.A. 9.0.1 | Id: 7/19

DVI A(TX)



Update notes: Rename the DVI_TX to DVI_A(TX).

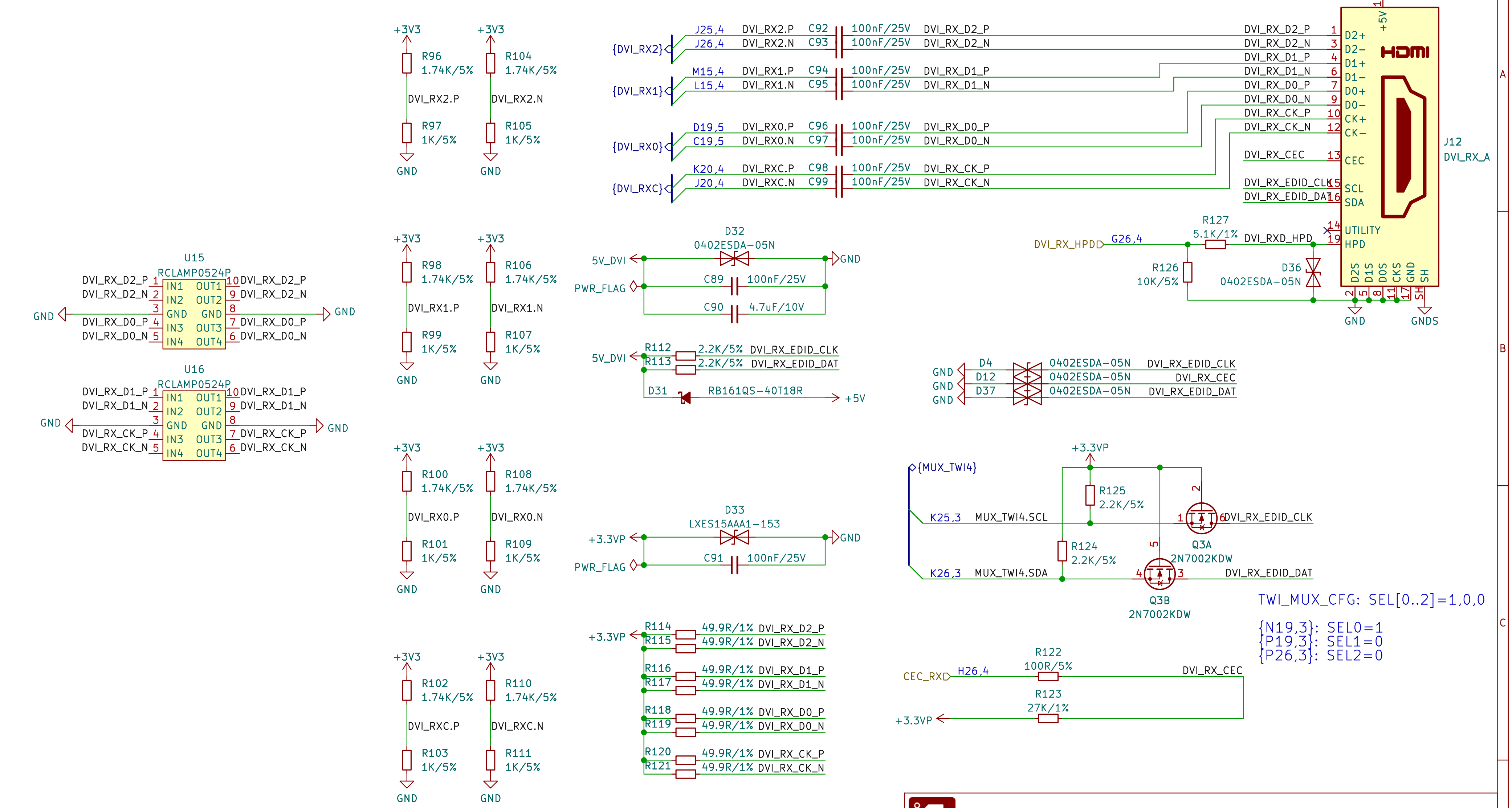
Shenzhen Sipeed Technology Co,Ltd.

Sheet: /FPGA_TMDS_A/
File: DVI_TX.kicad_sch

Title: TANG_MEGA-138K-Pro_Dock-4071f_Schematics

Size: A4	Date: 2025-03-25	**Rev: 1.5**
KiCad E.D.A. 9.0.1		Id: 8/19

DVI B(RX)



Update notes: Rename the DVI_RX to DVI_B(RX)

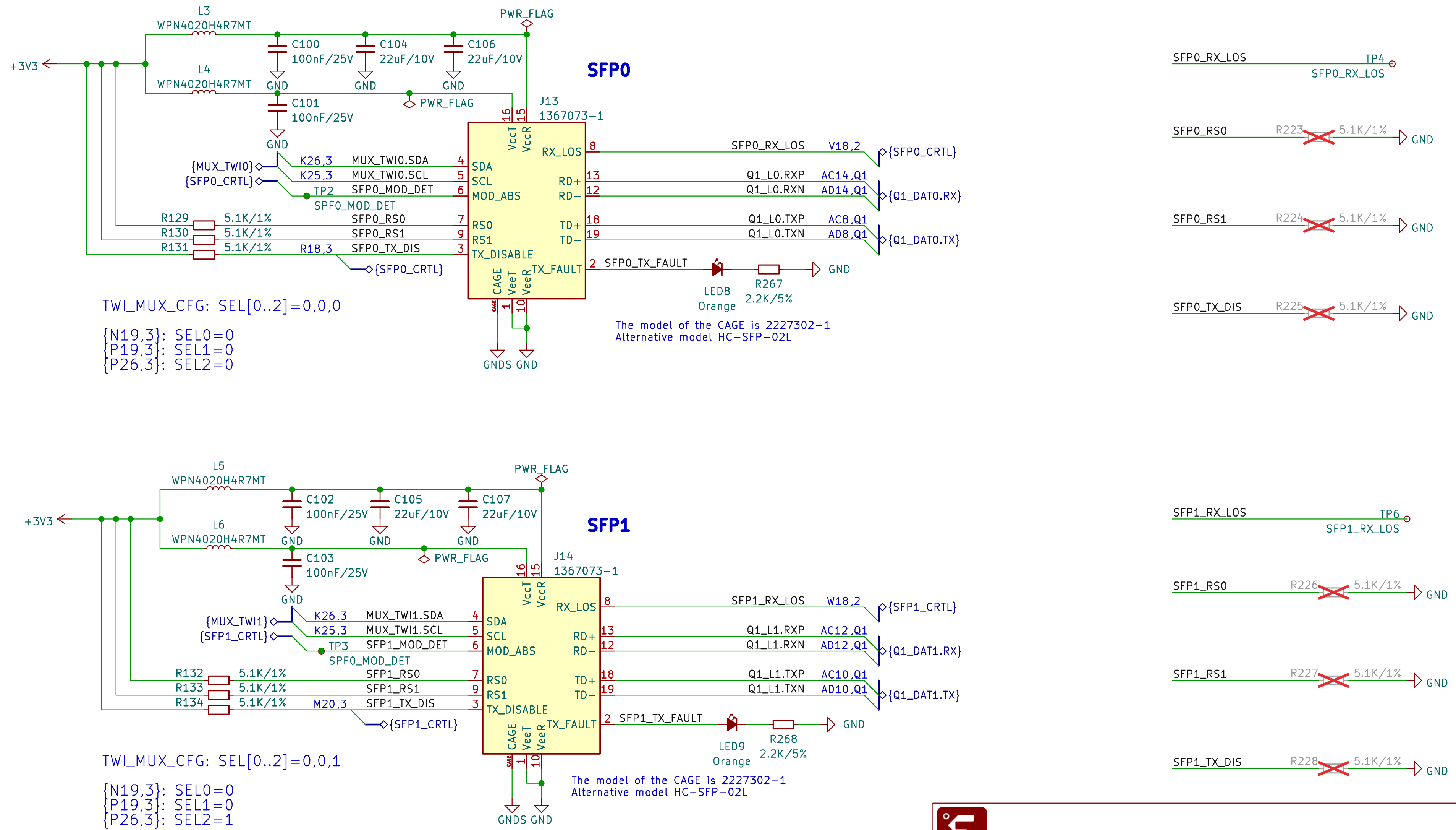
Shenzhen Sipeed Technology Co,Ltd.

Sheet: /FPGA_TMDS_B/
File: DVI_RX.kicad_sch

Title: TANG_MEGA-138K-Pro_Dock-4071f_Schematics

Size: A4	Date: 2025-03-25	**Rev: 1.5**
KiCad E.D.A. 9.0.1		Id: 9/19

SFP+



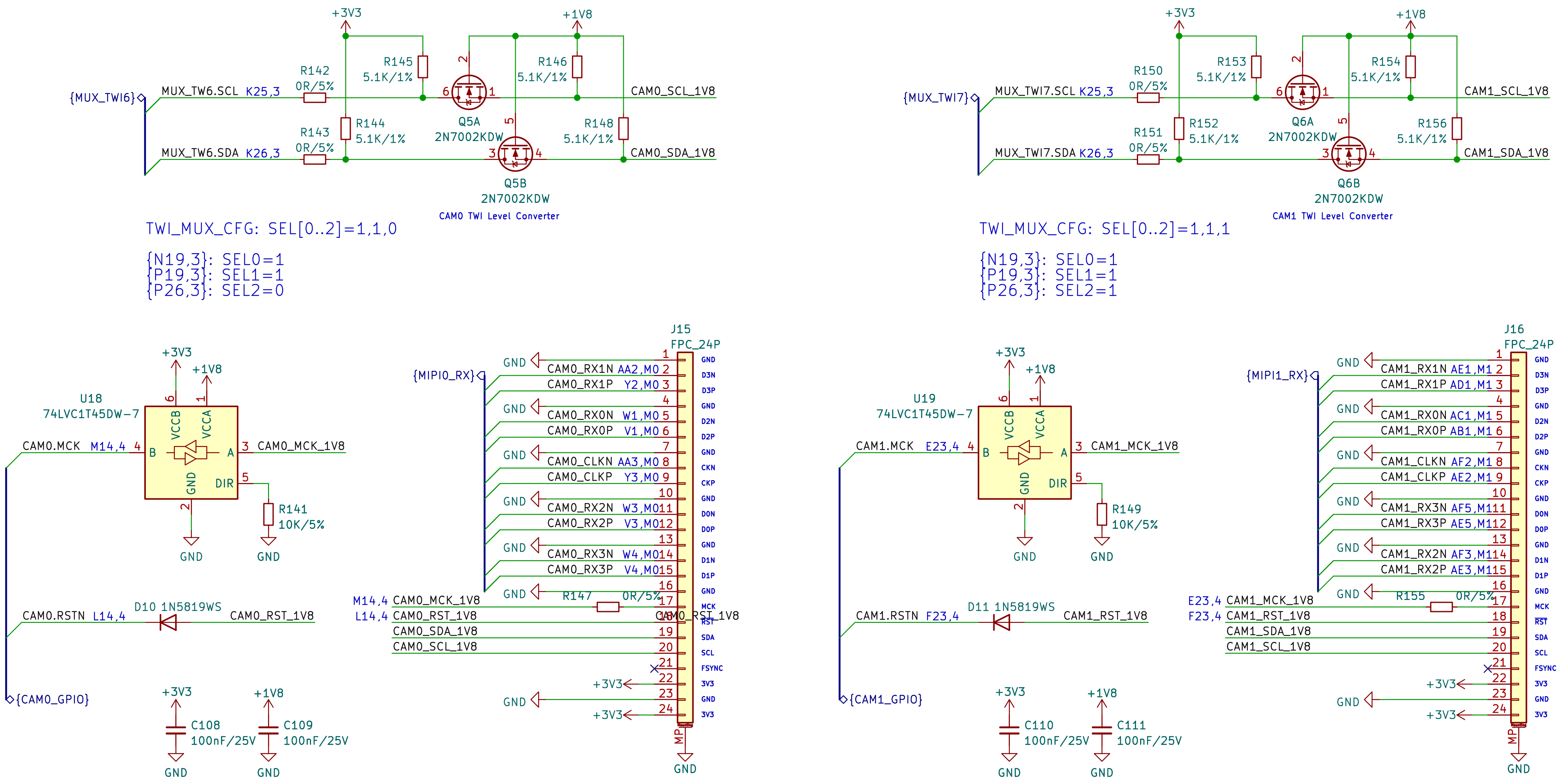
Shenzhen Sipeed Technology Co,Ltd.

Sheet: /FPGA_SFP+/
File: SFP-PLUS.kicad_sch

Title: TANG_MEGA-138K-Pro_Dock-4071f_Schematics

Size: A4	Date: 2025-03-25	**Rev: 1.5**
KiCad E.D.A. 9.0.1		Id: 10/19

MIPI RX



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Sheet: /FPGA_MIPI_RX/

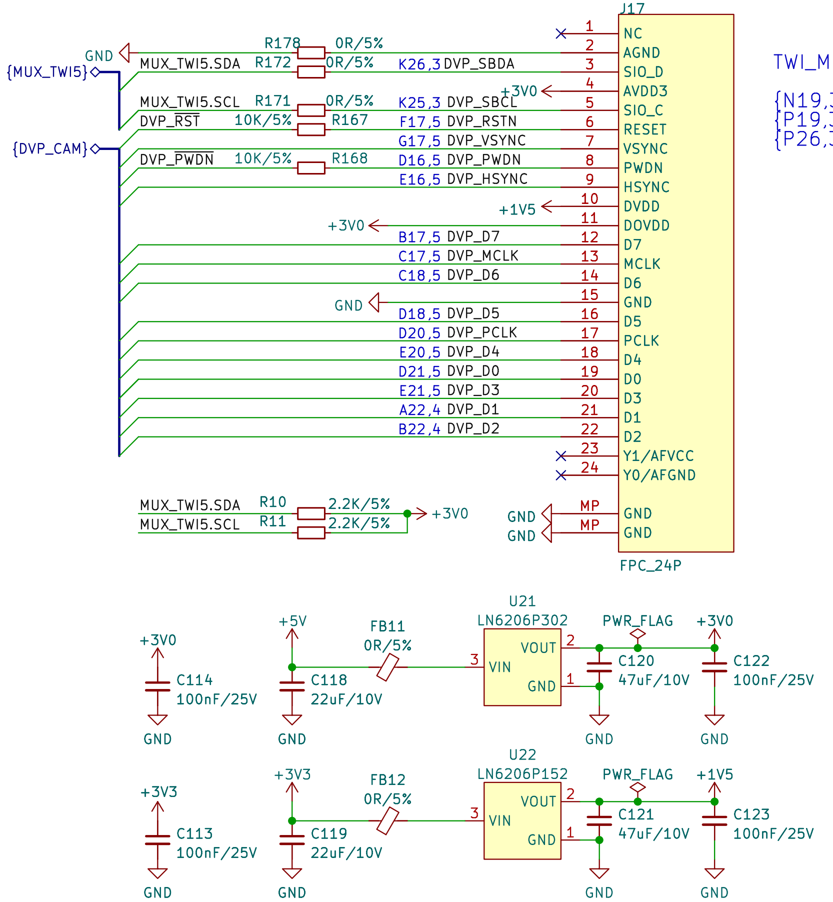
File: MIPI_RX.kicad_sch

Title: TANG_MEGA-138K-Pro_Dock-4071f_Schematics

Size: A4 Date: 2025-03-25 Rev: 1.5

KiCad E.D.A. 9.0.1 Id: 11/19

DVP-CAMERA



TWI_MUX_CFG: SEL[0..2]=1,0,1

{N19,3}: SEL0=1
{P19,3}: SEL1=0
{P26,3}: SEL2=1

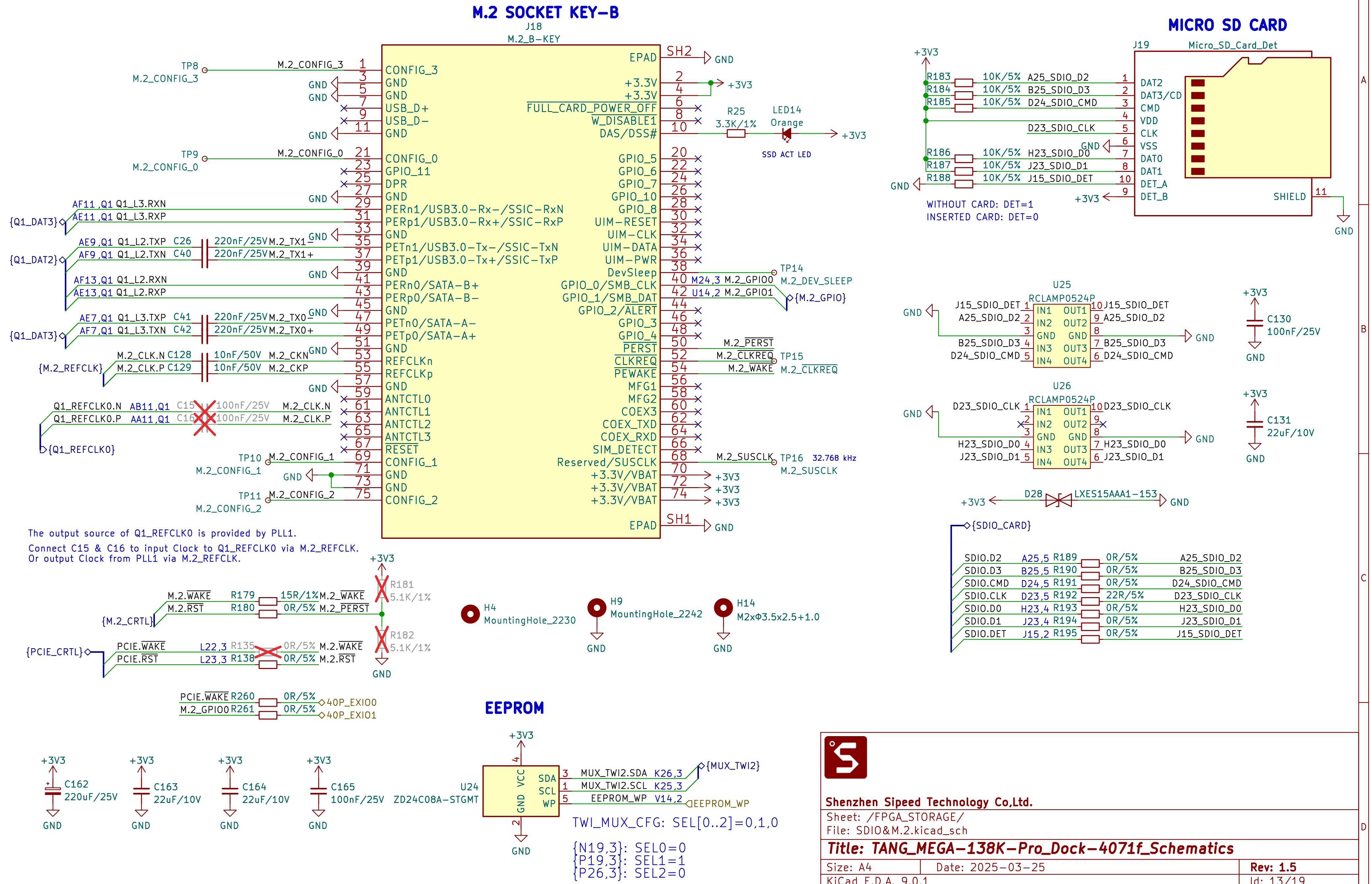
Shenzhen Sipeed Technology Co,Ltd.

Sheet: /DVP_CAMERA/
File: DVP.kicad_sch

Title: TANG_MEGA-138K-Pro_Dock-4071f_Schematics

Size: A4 Date: 2025-03-25 **Rev: 1.5**
KiCad E.D.A. 9.0.1 Id: 12/19

STORAGE



The output source of Q1_REFCLK0 is provided by PLL1.

Connect C15 & C16 to input Clock to Q1_REFCLK0 via M.2_REFCLK.
Or output Clock from PLL1 via M.2_REFCLK.

TWI_MUX_CFG: SEL[0..2]=0,1,0

{N19,3}: SEL0=0
{P19,3}: SEL1=1
{P26,3}: SEL2=0

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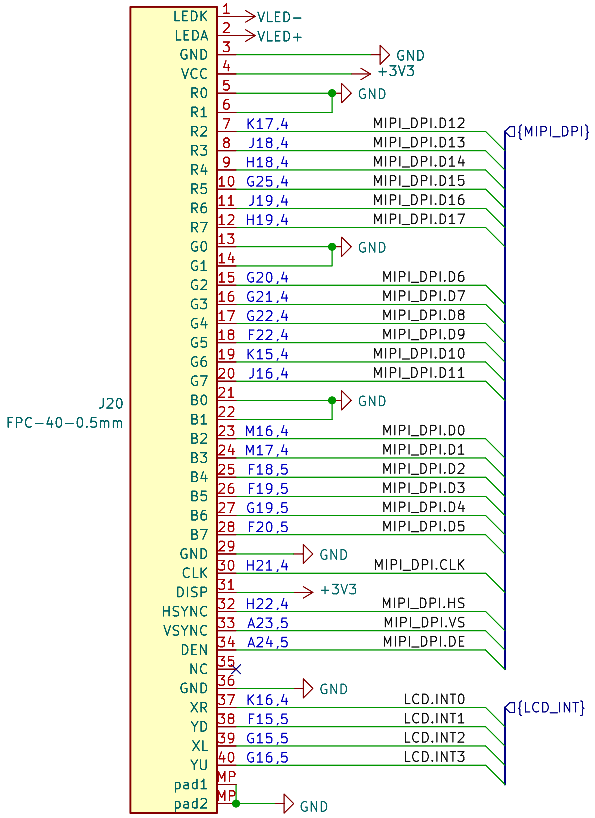
Sheet: /FPGA_STORAGE/
File: SDIO&M.2.kicad_sch

Title: TANG_MEGA-138K-Pro_Dock-4071f_Schematics

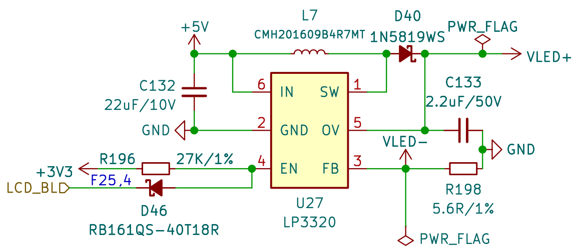
Size: A4	Date: 2025-03-25	Rev: 1.5
KiCad E.D.A. 9.0.1		Id: 13/19

FPGA-LCD

MIPI DPI CONN.



LCD BACKLIGHT



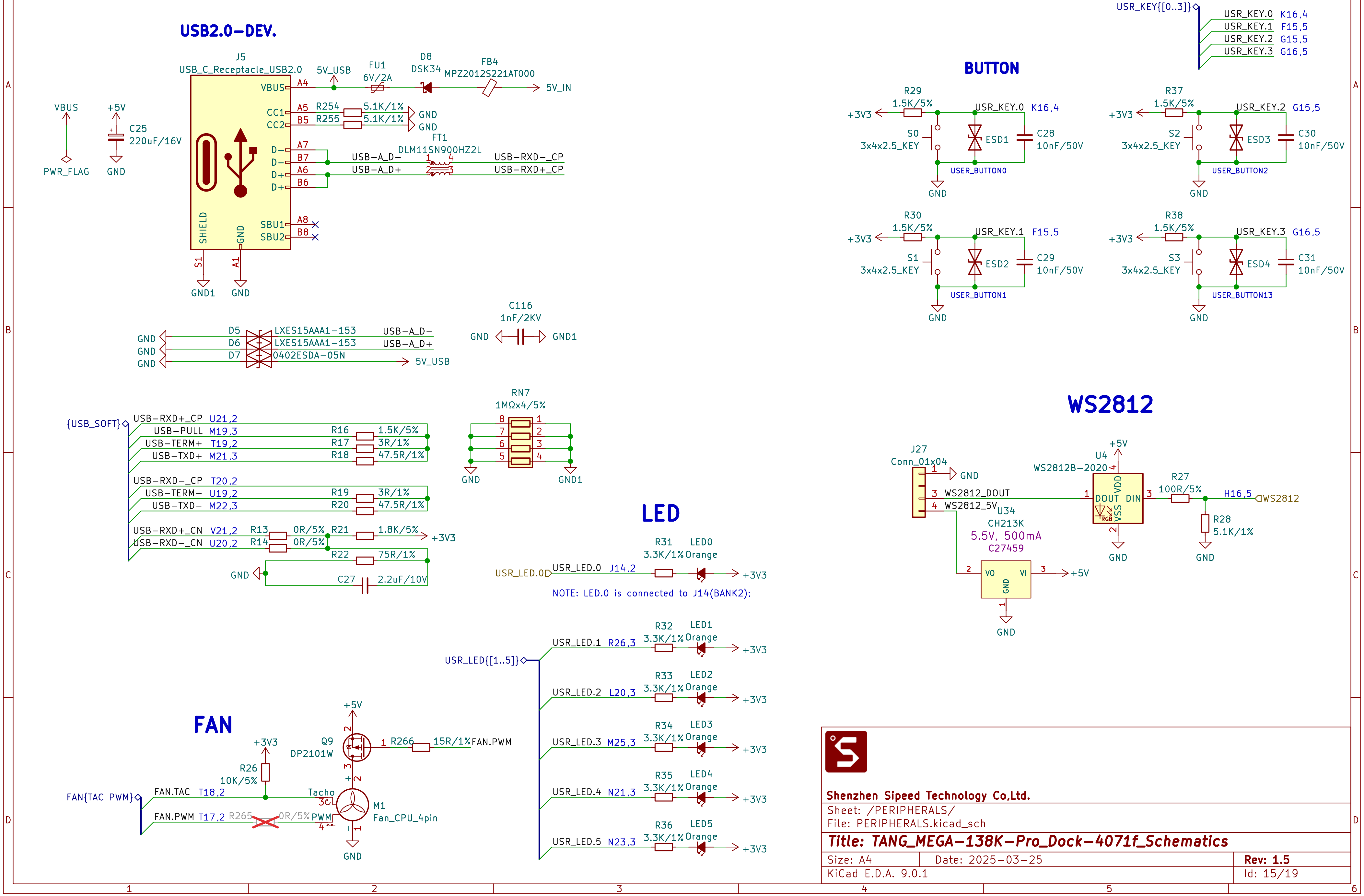
Shenzhen Sipeed Technology Co,Ltd.

Sheet: /FPGA_LCD/
File: LCD_DPI.kicad_sch

Title: TANG_MEGA-138K-Pro_Dock-4071f_Schematics

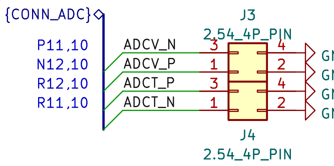
Size: A4	Date: 2025-03-25	**Rev: 1.5**
KiCad E.D.A. 9.0.1		Id: 14/19

PERIPHERALS

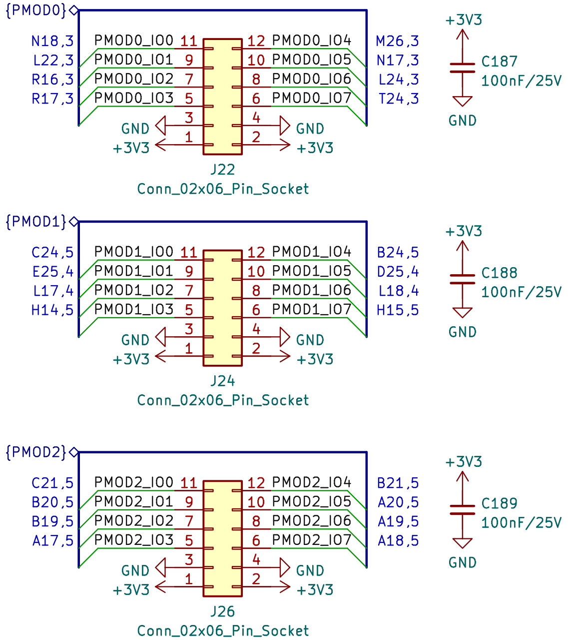


EX CONN.

ADC CONN.



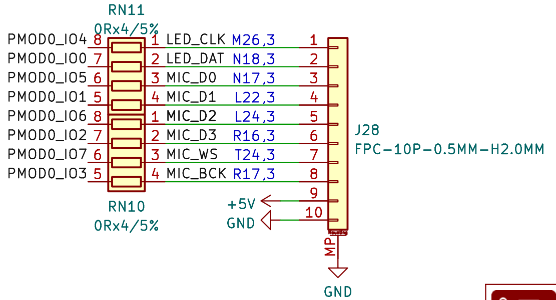
PMOD



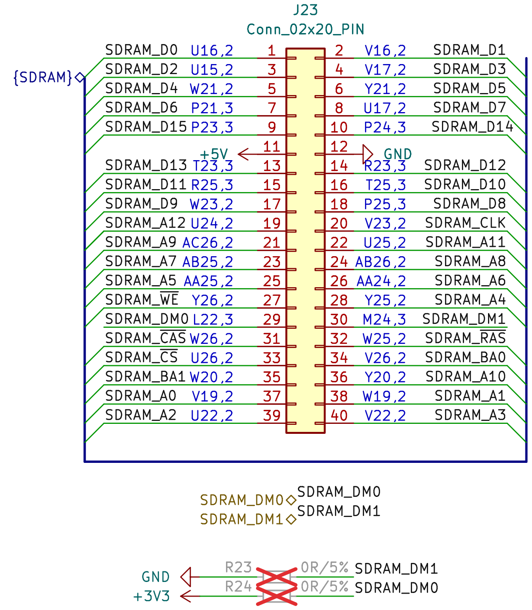
SSPI

PIN	FUCTION	PMOD0
N18	SSPI_SO	IO0
R16	SSPI_CLK	IO2
R17	SSPI_WP	IO3
N17	SSPI_SI	IO5
L24	SSPI_CS	IO6
T24	SSPI_HOLD	IO7

MIC-ARRAY CONN.



SDRAM CONN.



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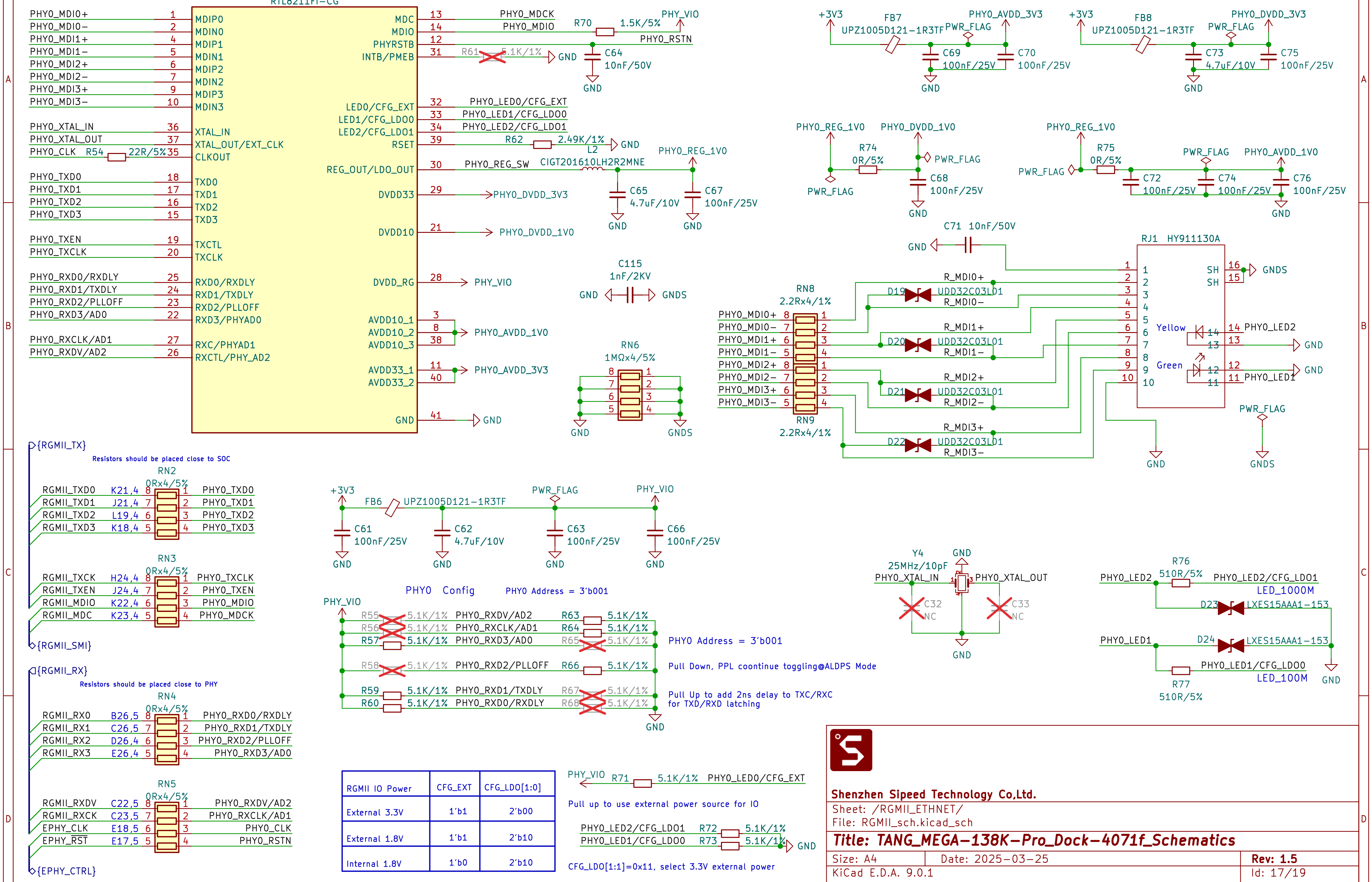
Sheet: /FPGA_EXT_CONN./
File: EX_CONN..kicad_sch

Title: TANG_MEGA-138K-Pro_Dock-4071f_Schematics

Size: A4 | Date: 2025-03-25 | Rev: 1.5
KiCad E.D.A. 9.0.1 | Id: 16/19

RGMII-ETHERNET



RGMII IO Power	CFG_EXT	CFG_LDO[1:0]
External 3.3V	1'b1	2'b00
External 1.8V	1'b1	2'b10
Internal 1.8V	1'b0	2'b10

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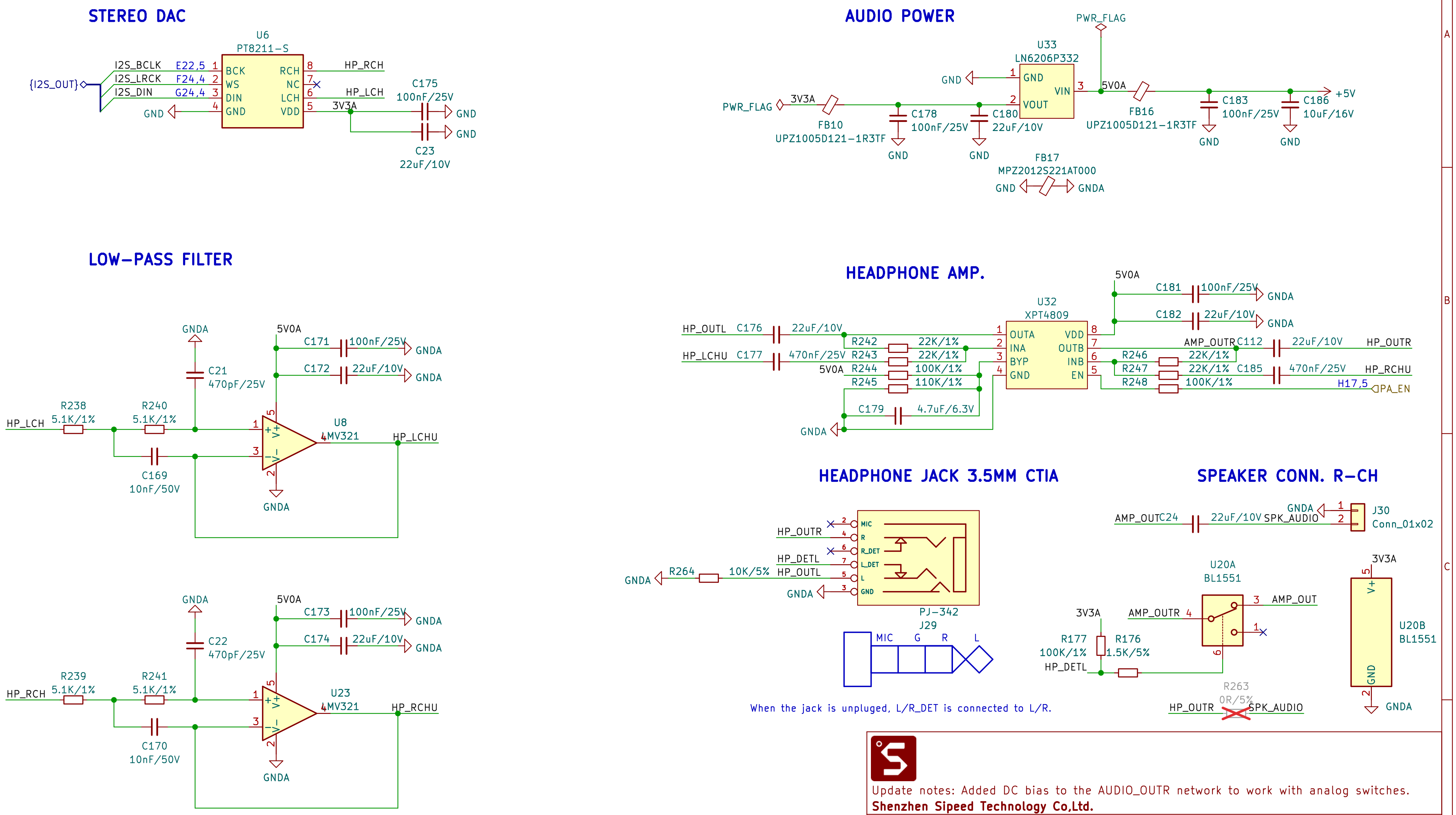
Sheet: /RGMII_ETHERNET/
File: RGMII_sch.kicad_sch

Title: TANG_MEGA-138K-Pro_Dock-4071f_Schematics

Size: A4 | Date: 2025-03-25 | **Rev: 1.5**

KiCad E.D.A. 9.0.1 | Id: 17/19

FPGA-Audio



Update notes: Added DC bias to the AUDIO_OUTR network to work with analog switches.

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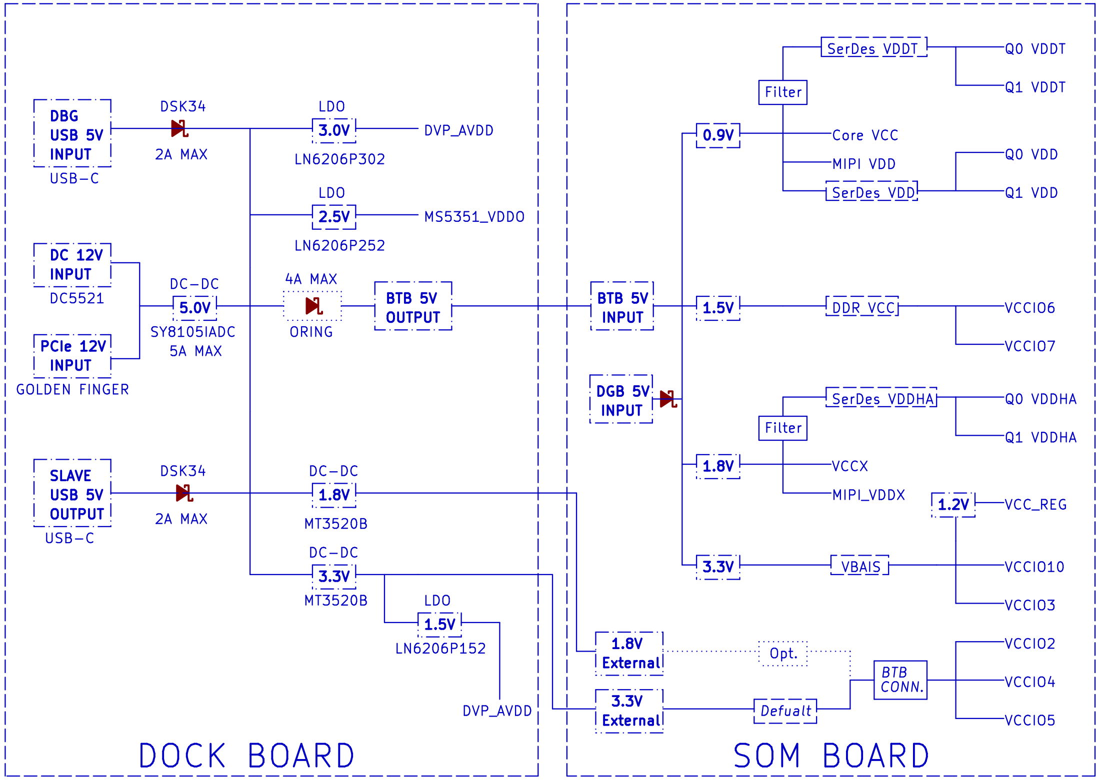
Sheet: /FPGA_AUDIO/
File: DAC+PA.kicad_sch

Title: TANG_MEGA-138K-Pro_Dock-4071f_Schematics

Size: A4 | Date: 2025-03-25 | **Rev: 1.5**

KiCad E.D.A. 9.0.1 | Id: 18/19



Update notes: Rename the DVI_TX_DDC DVI_A_DDC, DVI_RX_DDC to DVI_B_DDC.

Shenzhen Sipeed Technology Co,Ltd.

Sheet: /POWER_TREE/
File: PT.kicad_sch

Title: TANG_MEGA-138K-Pro_Dock-4071f_Schematics

Size: A4	Date: 2025-03-25	**Rev: 1.5**
KiCad E.D.A. 9.0.1		Id: 19/19